

Design and Deployment of a Highly Available, Multi-Tier Cloud-Native Application using the AWS Well-Architected Framework

VeriTrust – AI-Powered Fake News Detector

Submitted by

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Review: AWS Well-Architected Framework – 57/57 Questions Answered

Submission package distinguishes implemented AWS evidence from planned/target architecture.

1. Executive Summary

VeriTrust is an AI-powered fake-news detection application that combines machine-learning-based text classification with real-time web verification. The project uses Python, Streamlit, scikit-learn, TF-IDF and web-scraping components. The cloud case study applies the AWS Well-Architected Framework to design a highly available, multi-tier architecture.

The current implementation evidence confirms that the static VeriTrust frontend has been deployed to Amazon S3 and accessed successfully through the S3 website endpoint. The multi-tier target design adds Route 53, an Application Load Balancer, EC2 Auto Scaling across Availability Zones, Amazon RDS, VPC/security controls and CloudWatch. These components are explicitly marked as target/planned unless separately deployed.

Key outcomes

Area	Status
S3 static frontend	Implemented and verified
AWS WAFR review	57/57 questions answered
WAFR risk summary	16 High, 8 Medium at evidence capture
Multi-tier architecture	Target architecture documented
Public GitHub repository	Available
Infrastructure as Code	Not supplied / not currently implemented

2. Project Overview

VeriTrust provides users with an interface to submit news text or an article URL for analysis. The application performs text processing and machine-learning classification and can use live web verification to support the verdict.

Technology stack

Layer	Technology
Language	Python
ML/NLP	Scikit-learn, TF-IDF, NumPy, Pandas
Model serialization	Joblib
Web application	Streamlit
Web fetching	Requests, BeautifulSoup4
Visualization	Plotly
Cloud frontend	Amazon S3 static website
Source control	GitHub

ML result referenced by project documentation

The project documentation reports that the WELFake dataset contains 72,134 articles and that the Passive Aggressive Classifier achieved approximately 96.60% test accuracy, with Logistic Regression and Multinomial Naive Bayes also evaluated.

3. AWS Target Architecture

The target architecture separates presentation, application and data responsibilities. Amazon S3 hosts the static frontend. Route 53 provides DNS, the Application Load Balancer distributes requests, EC2 instances run the Python/Streamlit application across Availability Zones, and Amazon RDS provides managed relational storage in private subnets. VPC, IAM, security groups and CloudWatch provide supporting network, identity and observability controls.

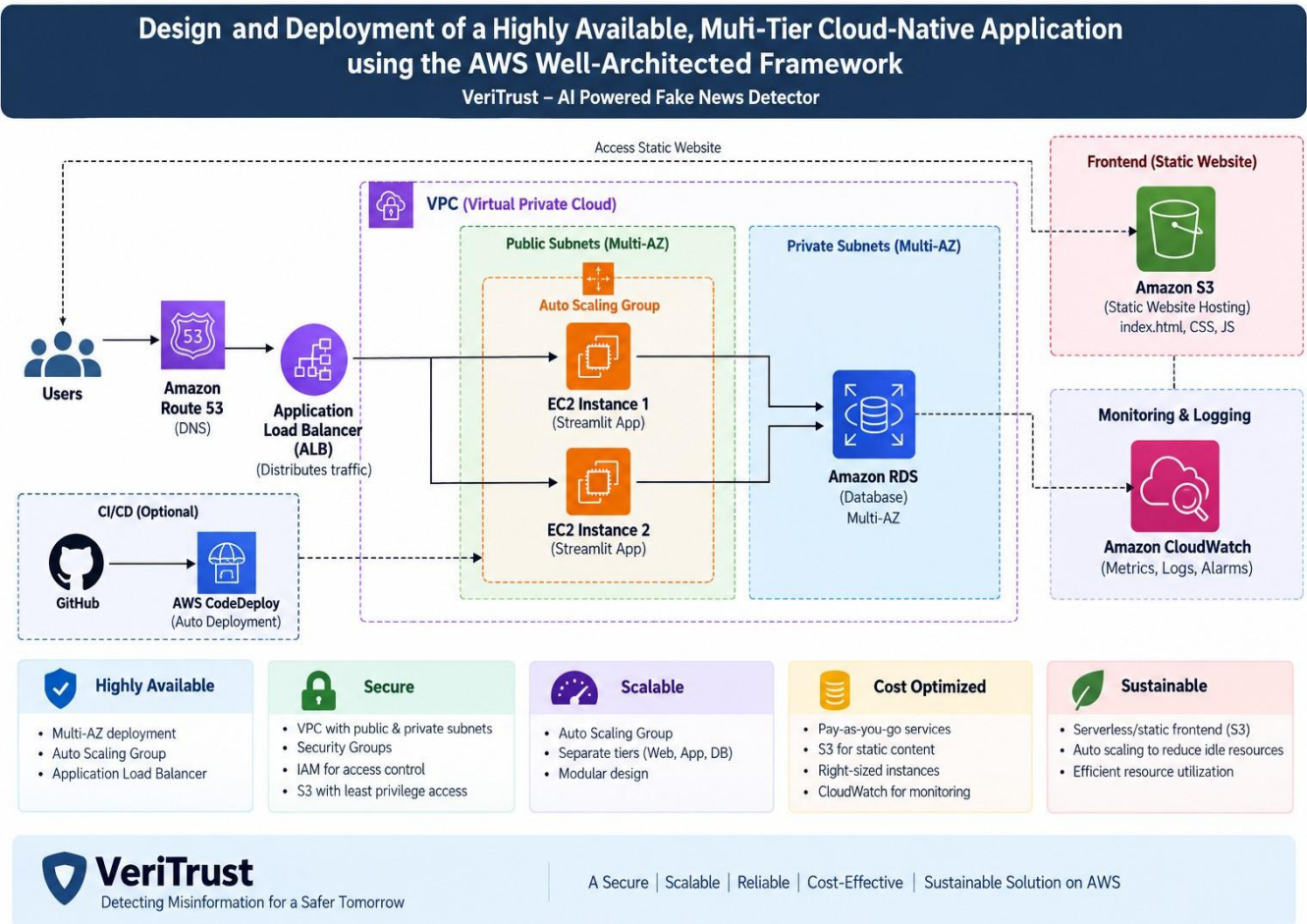


Figure 1. VeriTrust AWS multi-tier target architecture.

4. Multi-Tier and High-Availability Design

Tier	AWS component	Purpose
Presentation	Amazon S3	Static frontend assets: HTML, CSS and JavaScript
DNS	Amazon Route 53	Stable DNS name for the application
Traffic	Application Load Balancer	Distributes requests to healthy application instances
Application	Amazon EC2 + Auto Scaling	Runs Streamlit/Python application across multiple AZs
Data	Amazon RDS Multi-AZ	Managed relational data with failover capability
Network	Amazon VPC	Subnets, routing and isolation
Security	IAM + Security Groups	Identity, permissions and network access control
Monitoring	Amazon CloudWatch	Metrics, logs and alarms

High availability is achieved in the target design by distributing application compute across multiple Availability Zones, using ALB health checks and Auto Scaling replacement. The database tier is intended to use an RDS Multi-AZ configuration where supported by the selected engine.

5. AWS Well-Architected Framework Review

The AWS Well-Architected Tool review for VeriTrust contains all 57 questions. The evidence screenshot captured from the AWS console shows 57/57 questions answered. The six-pillar breakdown is shown below.

Pillar	Questions
Operational Excellence	11/11
Security	11/11
Reliability	13/13
Performance Efficiency	5/5
Cost Optimization	11/11
Sustainability	6/6
Total	57/57

Current risk summary

The AWS console evidence captured after completing the review shows 16 High-risk and 8 Medium-risk items. These are treated as improvement areas rather than evidence that all target controls are already deployed.

6. Six-Pillar Alignment

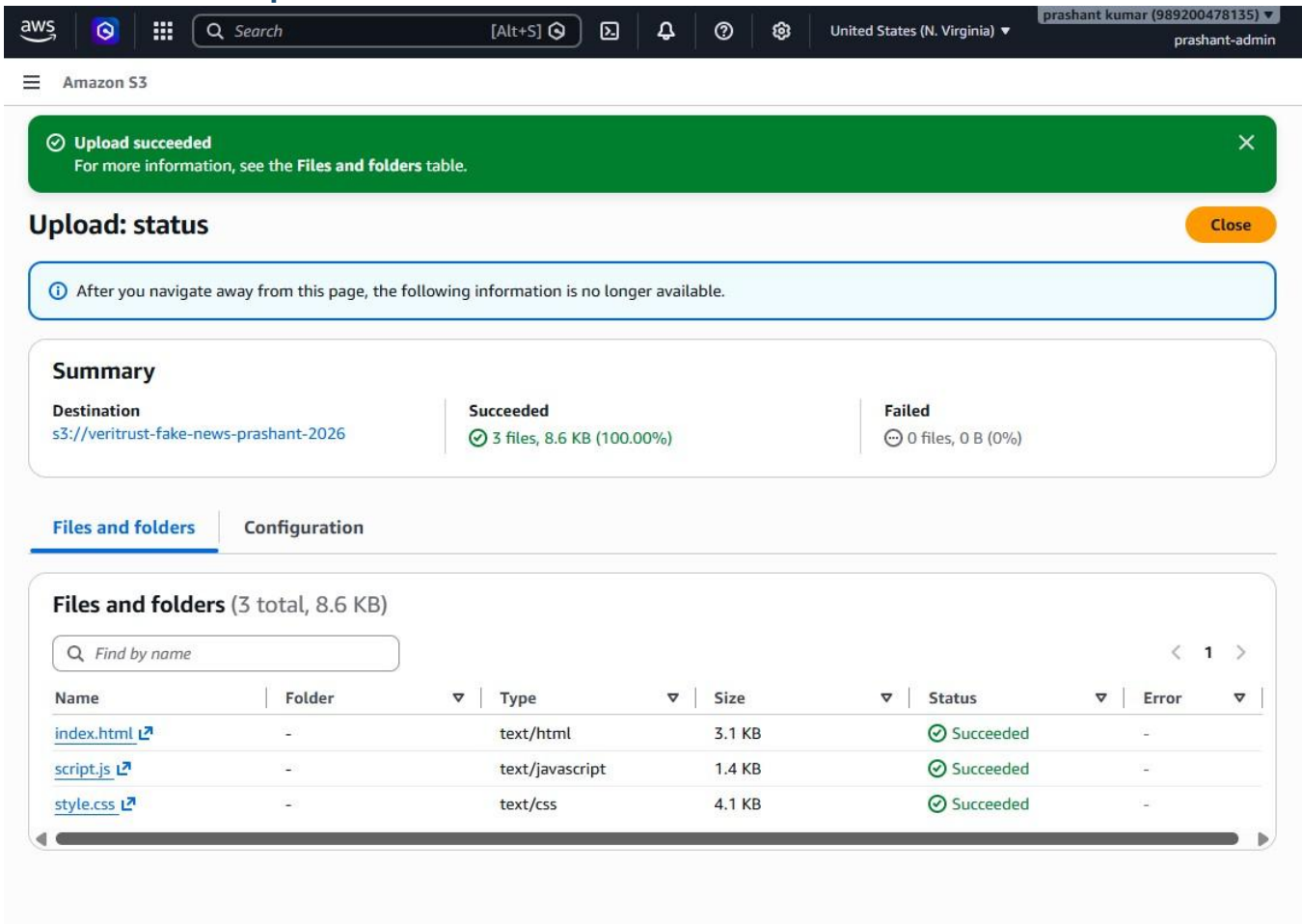
Pillar	VeriTrust design response
Operational Excellence	Version control, deployment procedures, monitoring and continuous improvement.
Security	Least-privilege IAM, VPC segmentation, security groups, encryption and controlled access.
Reliability	Multi-AZ application tier, ALB health checks, Auto Scaling, backups and managed database.
Performance Efficiency	Right-sized compute, horizontal scaling, load distribution and performance monitoring.
Cost Optimization	Right-sizing, demand-based scaling, cost monitoring and removal of unused resources.
Sustainability	Managed services, dynamic scaling, efficient utilization and data/resource lifecycle management.

The review answers were prepared against the target architecture requested for the case study. For final academic submission, practices should be considered implemented only after the corresponding AWS configuration or operational process is actually deployed and validated.

7. Deployment Evidence

The following evidence demonstrates the portion of the architecture that has actually been deployed during this case study: the Amazon S3 static frontend and the Well-Architected review workflow.

7.1 S3 frontend upload



Upload succeeded
For more information, see the [Files and folders](#) table.

Upload: status Close

After you navigate away from this page, the following information is no longer available.

Summary

Destination s3://veritrust-fake-news-prashant-2026	Succeeded 3 files, 8.6 KB (100.00%)	Failed 0 files, 0 B (0%)
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Files and folders | Configuration

Files and folders (3 total, 8.6 KB)

Find by name

Name	Folder	Type	Size	Status	Error
index.html	-	text/html	3.1 KB	✓ Succeeded	-
script.js	-	text/javascript	1.4 KB	✓ Succeeded	-
style.css	-	text/css	4.1 KB	✓ Succeeded	-

Figure 2. Successful upload of the three static frontend files to the S3 bucket.

7.2 Static website hosting configuration

Amazon S3 > Buckets > veritrust-fake-news-prashant-2026 > Edit static website hosting

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting

☐ Disable

☒ Enable

Hosting type

☒ Host a static website

Use the bucket endpoint as the web address. [Learn more](#)

☐ Redirect requests for an object

Redirect requests to another bucket or domain. [Learn more](#)

For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)

Index document

Specify the home or default page of the website.

index.html

Error document - optional

This is returned when an error occurs.

error.html

Redirection rules – optional

Redirection rules, written in JSON, automatically redirect webpage requests for specific content. [Learn more](#)

1

Figure 3. Amazon S3 static website hosting configuration with index.html.

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7.3 Working deployed frontend

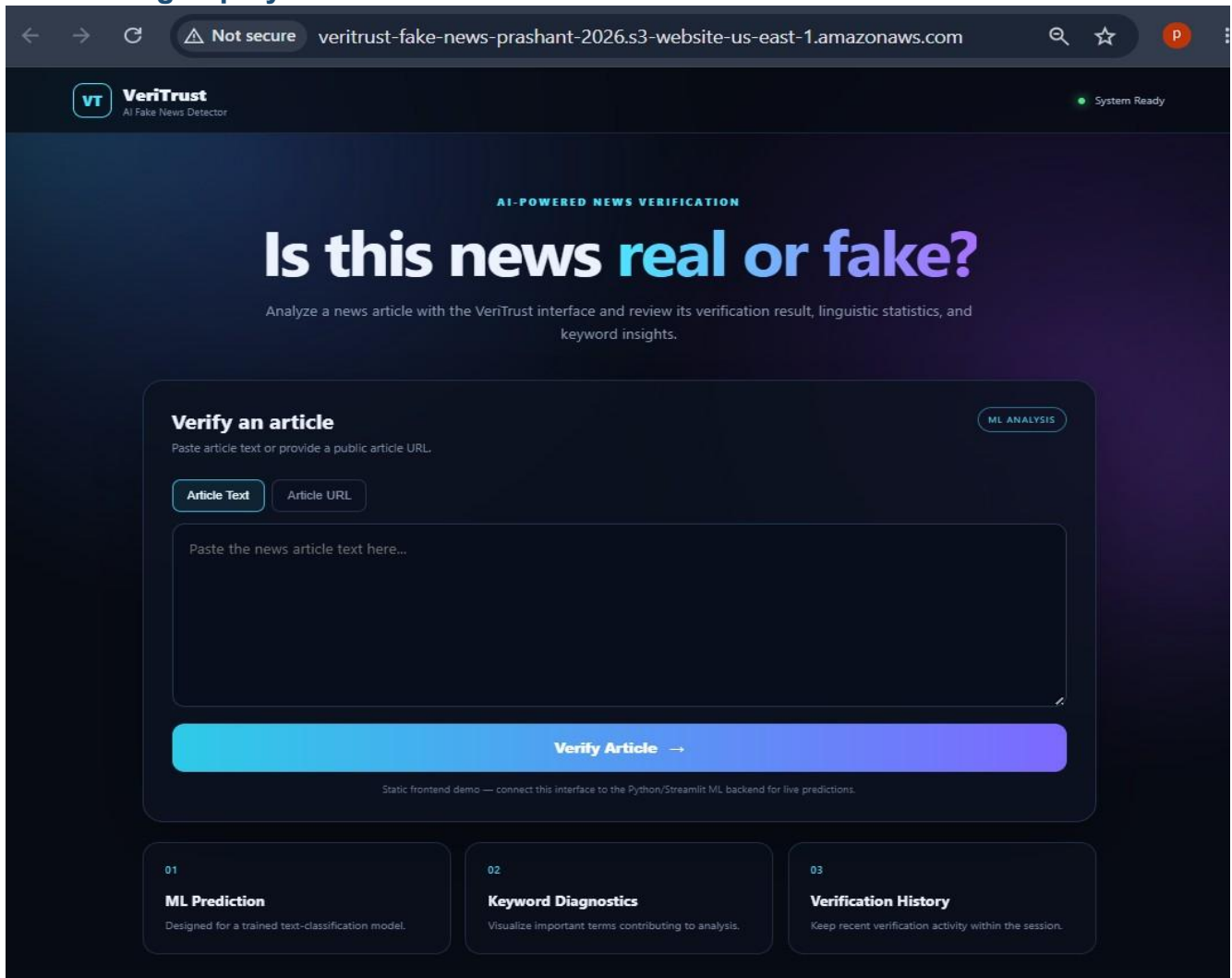


Figure 4. VeriTrust frontend successfully served from the S3 website endpoint.

8. Well-Architected Tool Evidence

8.1 57/57 questions answered

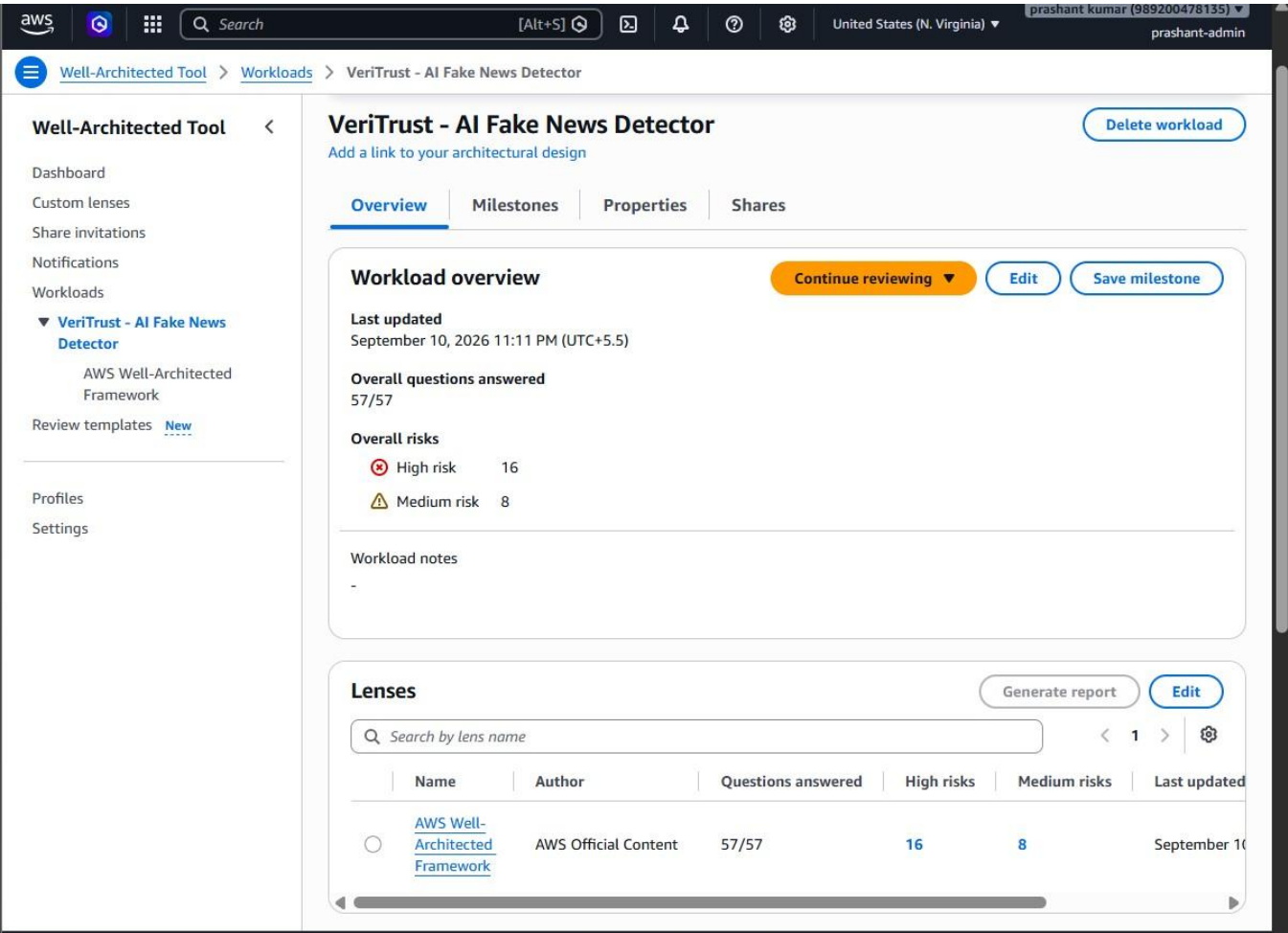


Figure 5. AWS Well-Architected Tool showing 57/57 questions answered.

8.2 Milestone saved

The screenshot displays the AWS Well-Architected Tool interface. At the top, the navigation bar includes the AWS logo, a search bar, and user information for 'prashant kumar (989200478135)' in 'United States (N. Virginia)'. The breadcrumb trail shows 'Well-Architected Tool > Workloads > VeriTrust - AI Fake News Detector'. On the left sidebar, the 'VeriTrust - AI Fake News Detector' workload is selected under the 'Workloads' section. The main content area features two notification banners: a red one stating 'Milestone with this name already exists for this workload' and a green one stating 'Successfully saved milestone'. Below these, the workload title 'VeriTrust - AI Fake News Detector' is shown with a 'Delete workload' button. The 'Overview' tab is active, displaying the 'Workload overview' section. This section includes a 'Continue reviewing' button, an 'Edit' button, and a 'Save milestone' button. It also shows the last updated time as 'September 10, 2026 11:10 PM (UTC+5.5)', overall questions answered as '57/57', and overall risks with 16 high-risk items.

Figure 6. Successful Well-Architected milestone save.

9. Implementation Roadmap

Phase	Activity	Status
1	S3 static frontend deployment	Completed

2	VPC, subnets, route tables and security groups	Target / next
3	Deploy Python/Streamlit application to EC2	Target / next
4	Launch Template + Auto Scaling across two AZs	Target / next
5	Application Load Balancer + health checks	Target / next
6	RDS in private subnets	Target / next
7	CloudWatch metrics, logs and alarms	Target / next
8	Validate scaling, failure handling and availability	Target / next
9	Revalidate WAFR answers against actual deployment	Target / next
10	Download official AWS WAFR generated report	Pending

The current case study therefore presents the S3 deployment as completed evidence and the remaining multi-tier components as the target architecture required to fully realize the assignment title.

10. Submission Checklist

Requirement	Included
Case Study Report (PDF)	Yes
Source Code / Application Files	Static frontend included + public GitHub source link
AWS Architecture Diagram	Yes
Infrastructure / Deployment Files	Deployment notes included; no IaC supplied
Deployment Screenshots / Evidence	Yes
README with setup/deployment/testing instructions	Yes
Public GitHub repository link	https://github.com/Prashantjativ563/Fake-News-Detection

Deployment endpoint

<http://veritrust-fake-news-prashant-2026.s3-website-us-east-1.amazonaws.com>

Note: direct S3 static website hosting is HTTP. HTTPS can be added later with a CDN/edge layer such as CloudFront.

Conclusion

This case study establishes VeriTrust as a Well-Architected, multi-tier cloud design while preserving the already deployed S3 frontend. The AWS Well-Architected review has been completed at 57/57 questions, and the evidence package records both the deployed S3 implementation and the target high-availability architecture.